

Byte Pair Encoding (BPE)

Require: K

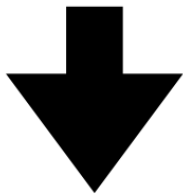
▷ Vocabulary size

- 1: $V \leftarrow$ all characters in the training data
- 2: **while** $|V| \leq K$ **do**
- 3: Pre-tokenize and tokenize the data by repeatedly taking the longest matching prefix
- 4: Count the frequency of all adjacent token pairs in the data
- 5: Let $\langle l, r \rangle$ be the most frequent adjacent token pair
- 6: $V \leftarrow V \cup \{lr\}$ ▷ Add the merged token as a new token
- 7: **end while**
- 8: **return** V

Simple Execution

$K = 10$

Word	Frequency
hug	10
pug	5
pun	12
bun	4
hugs	5



$\mathcal{V} = \{b, g, h, n, p, s, u\}$

$K = 10$
 $\mathcal{V} = \{b, g, h, n, p, s, u\}$

Word	Frequency
h+u+g	10
p+u+g	5
p+u+n	12
b+u+n	4
h+u+g+s	5

Pair	Frequency
u+g	20
p+u	17
u+n	16
h+u	15
g+s	5
⋮	

$$K = 10$$

$$\mathcal{V} = \{b, g, h, n, p, s, u, ug\}$$

Word	Frequency
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h+ug	10
------	----

p+ug	5
------	---

p+u+n	12
-------	----

b+u+n	4
-------	---

h+ug+s	5
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Pair	Frequency
------	-----------

u+n	16
-----	----

h+ug	15
------	----

p+u	12
-----	----

p+ug	5
------	---

ug+s	5
------	---

⋮

$$K = 10$$

$$\mathcal{V} = \{b, g, h, n, p, s, u, ug, un, hug\}$$

Word	Frequency
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hug	10
-----	----

p+ug	5
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p+un	12
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b+un	4
------	---

hug+s	5
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Subword Tokenization

- To avoid unknown words, we must add all possible characters/symbols
- But, there are ~138K unicode symbols
- Instead, use bytes!
 - LLMs often use BPE with some rules to prevent certain types of merges
 - Modest-sized LLMs often have vocabulary sizes of 32-64K
 - With the big proprietary models, it is often ~100k–256k tokens